Algebra & Geometry Practice Packet

Solving Equations · Quadratics · Radical Functions · Geometry

Name: _____ **Date:** _____

This packet has three difficulty levels throughout: **Easy**, **Medium**, and **Hard**. Each topic opens with a worked *Model*, followed by similar practice. Full solutions are in the Answer Key at the end.

1. Solving Linear Equations

Easy — One-Step Equations

Model. Solve $x + 7 = 12$.

$$\begin{aligned} x + 7 &= 12 && \text{original equation} \\ x &= 12 - 7 && \text{subtract 7 from both sides} \\ x &= 5 && \text{simplify} \end{aligned}$$

Solve each: (a) $x - 4 = 9$ (b) $5x = 35$ (c) $\dfrac{x}{3} = 6$ (d) $x + 12 = 20$

Medium — Two-Step Equations

Model. Solve $2x + 3 = 11$.

$$\begin{aligned} 2x + 3 &= 11 && \text{original equation} \\ 2x &= 8 && \text{subtract 3 from both sides} \\ x &= 4 && \text{divide both sides by 2} \end{aligned}$$

Solve each: (a) $3x - 5 = 16$ (b) $4x + 7 = 31$ (c) $\dfrac{x}{2} - 3 = 5$ (d) $-2x + 9 = 1$

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Hard — Multi-Step Equations

Model. Solve $3(x-2)+4=2x+5$.

$$\begin{aligned} 3(x-2)+4 &= 2x+5 && \text{original equation}\\ 3x-6+4 &= 2x+5 && \text{distribute the 3}\\ 3x-2 &= 2x+5 && \text{combine like terms}\\ x-2 &= 5 && \text{subtract } 2x \text{ from both sides}\\ x &= 7 && \text{add 2 to both sides} \end{aligned}$$

Solve each:

(a) $2(x+3)=x+10$

(b) $5x-2=3x+8$

(c) $4(x-1)+2=2(x+3)$

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(d) $3x + 2(x - 1) = 18$

2. Solving Quadratics by Factoring (Zero-Product Property)

Zero-Product Property. If $A \cdot B = 0$, then $A = 0$ *or* $B = 0$. So a quadratic in factored form is solved by setting each factor equal to 0.

Easy — Already Factored

Model. Solve $(x + 1)(x - 2) = 0$.

$$\begin{aligned} (x+1)(x-2) &= 0 && \text{original equation} \\ x+1=0 \ \text{ or } \ x-2 &= 0 && \text{zero-product property} \\ x=-1 \ \text{ or } \ x &= 2 && \text{solve each factor} \end{aligned}$$

Solve each: (a) $(x-3)(x+5) = 0$ (b) $(x+4)(x-6) = 0$ (c) $x(x-7) = 0$ (d) $(2x-1)(x+3) = 0$

Medium — Factor First (leading coefficient $= 1$)

Model. Solve $x^2 + 5x + 6 = 0$.

$$\begin{aligned} x^2+5x+6 &= 0 && \text{original equation} \\ (x+2)(x+3) &= 0 && \text{factor: } 2\cdot 3=6,\ 2+3=5 \\ x+2=0 \ \text{ or } \ x+3 &= 0 && \text{zero-product property} \\ x=-2 \ \text{ or } \ x &= -3 && \text{solve each factor} \end{aligned}$$

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Solve each: (a) $x^2+7x+12=0$ (b) $x^2-5x+6=0$ (c) $x^2-x-12=0$ (d) $x^2+2x-15=0$

Hard — Factor First (leading coefficient $\neq 1$)

Model. Solve $2x^2+7x+3=0$.

$$\begin{aligned} 2x^2+7x+3&=0 && \text{original equation}\\ (2x+1)(x+3)&=0 && \text{factor (product 6, sum 7)}\\ 2x+1=0 \ \text{ or } \ x+3&=0 && \text{zero-product property}\\ x=-\tfrac{1}{2} \ \text{ or } \ x&=-3 && \text{solve each factor} \end{aligned}$$

Solve each: (a) $3x^2-10x+8=0$ (b) $2x^2-5x-3=0$ (c) $6x^2+x-2=0$ (d) $4x^2-4x-3=0$

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3. Solving Quadratics by the Quadratic Formula

Quadratic Formula. For $ax^2 + bx + c = 0 \quad (a \neq 0)$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The value under the root, $b^2 - 4ac$, is the *discriminant*: positive $\Rightarrow$ two solutions, zero $\Rightarrow$ one, negative $\Rightarrow$ no real solution.

Model. Solve $x^2 - 5x + 6 = 0 \quad (a = 1,\ b = -5,\ c = 6)$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \text{quadratic formula}$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(6)}}{2(1)} \qquad \text{substitute } a,\ b,\ c$$

$$x = \frac{5 \pm \sqrt{25 - 24}}{2} = \frac{5 \pm 1}{2} \qquad \text{simplify under the root}$$

$$x = 3 \quad \text{or} \quad x = 2 \qquad \text{the two solutions}$$

Solve each with the quadratic formula:

(a) $x^2 + 3x - 10 = 0$

(b) $x^2 - 2x - 8 = 0$

(c) $2x^2 + 5x - 3 = 0$

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(d) $x^2 - 4x + 1 = 0$ *(leaves a radical)*

4. Radical Functions: Domain & Range

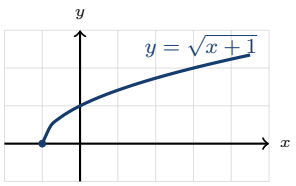
Square-Root Functions

For a square root $y = \sqrt{(\text{radicand})}$, the radicand must be ≥ 0 (you cannot take the square root of a negative). Solve radicand ≥ 0 to get the domain; the output of a square root is always ≥ 0.

Model. Find the domain and range of $y = \sqrt{x+1}$.

$$x + 1 \geq 0 \qquad \text{the radicand cannot be negative}$$
$$x \geq -1 \qquad \text{subtract 1 from both sides}$$

Domain: $x \geq -1,\ [-1, \infty)$. Range: $y \geq 0,\ [0, \infty)$ (a square root is never negative).



State the domain and range: (a) $y = \sqrt{x-3}$ (b) $y = \sqrt{x+5}$ (c) $y = \sqrt{2x-4}$ (d) $y = \sqrt{x} + 2$

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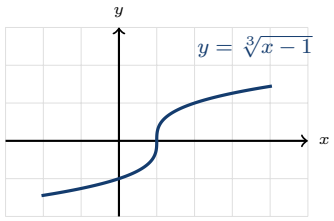
Cube-Root Functions

A cube root $y = \sqrt[3]{(\text{radicand})}$ accepts **every** real number (negatives are fine, since $\sqrt[3]{-8} = -2$). So both the domain and range are all real numbers.

Model. Find the domain and range of $y = \sqrt[3]{x-1}$.

$x - 1$: no restriction — a cube root is defined for every real number (e.g. $\sqrt[3]{-8} = -2$)

Domain: all reals $(-\infty, \infty)$. Range: all reals $(-\infty, \infty)$.



State the domain and range: (a) $y = \sqrt[3]{x+4}$ (b) $y = \sqrt[3]{x} - 2$ (c) $y = \sqrt[3]{2x-6}$ (d) $y = \sqrt[3]{x-1} + 3$

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5. Geometry

Sum of Interior Angles of a Polygon

The interior angles of a polygon with n sides add up to

$$(n-2)\cdot 180^\circ.$$

Model. Find the interior-angle sum of a pentagon ($n=5$).

$$\begin{aligned} \text{sum} &= (n-2)\cdot 180^\circ && \text{polygon angle-sum formula}\\ &= (5-2)\cdot 180^\circ && \text{substitute } n=5\\ &= 3\cdot 180^\circ = 540^\circ && \text{simplify} \end{aligned}$$

Find the interior-angle sum: (a) hexagon ($n=6$) (b) octagon ($n=8$) (c) quadrilateral ($n=4$) (d) decagon ($n=10$)

Classifying a Quadrilateral from Four Vertices

Plot the four points, then use the **distance formula** for side lengths and **slopes** for direction:

- Both pairs of opposite sides parallel ⇒ **parallelogram**.
- Parallelogram + all four sides equal ⇒ **rhombus**.
- Parallelogram + right angles ⇒ **rectangle**.
- All four sides equal *and* right angles ⇒ **square**.

Model. Classify $A(0,0)$, $B(4,0)$, $C(4,3)$, $D(0,3)$.

$$\begin{aligned} AB = DC = 4,\quad BC = AD = 3 && \text{side lengths (distance formula)}\\ AB \text{ horizontal, } BC \text{ vertical} && \text{so consecutive sides are perpendicular}\\ \Rightarrow \textbf{rectangle} && \text{has right angles, but } 4\neq 3 \text{ so not a square} \end{aligned}$$

Classify each quadrilateral $ABCD$:

(a) $A(0,0)$, $B(3,0)$, $C(3,3)$, $D(0,3)$

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(b) $A(0,0)$, $B(5,0)$, $C(5,2)$, $D(0,2)$

(c) $A(3,0)$, $B(0,2)$, $C(-3,0)$, $D(0,-2)$

(d) $A(0,0)$, $B(4,0)$, $C(6,3)$, $D(2,3)$

Surface Area of a Cylinder

A cylinder with radius r and height h has surface area

$$SA = 2\pi r^2 + 2\pi rh \qquad \text{(two circular ends + the side).}$$

Model. Find the surface area for $r = 3,\ h = 5$.

$$\begin{aligned} SA &= 2\pi r^2 + 2\pi rh && \text{surface-area formula} \\ &= 2\pi(3)^2 + 2\pi(3)(5) && \text{substitute } r = 3,\ h = 5 \\ &= 18\pi + 30\pi && \text{simplify each term} \\ &= 48\pi \approx 150.8 && \text{combine (square units)} \end{aligned}$$

Find the surface area (leave in terms of π, then round to one decimal): (a) $r = 2,\ h = 6$ (b) $r = 5,\ h = 10$ (c) $r = 4,\ h = 7$ (d) $r = 1,\ h = 8$

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6. Answer Key

1. Solving Linear Equations

Easy: (a) $x = 13$ (b) $x = 7$ (c) $x = 18$ (d) $x = 8$.
Medium: (a) $x = 7$ (b) $x = 6$ (c) $x = 16$ (d) $x = 4$.
Hard: (a) $2x + 6 = x + 10 \Rightarrow x = 4$ (b) $2x = 10 \Rightarrow x = 5$ (c) $4x - 2 = 2x + 6 \Rightarrow x = 4$ (d) $5x - 2 = 18 \Rightarrow x = 4$.

2. Quadratics by Factoring

Easy: (a) $x = 3, -5$ (b) $x = -4, 6$ (c) $x = 0, 7$ (d) $x = \frac{1}{2}, -3$.
Medium: (a) $(x+3)(x+4) \Rightarrow x = -3, -4$ (b) $(x-2)(x-3) \Rightarrow x = 2, 3$ (c) $(x-4)(x+3) \Rightarrow x = 4, -3$ (d) $(x+5)(x-3) \Rightarrow x = -5, 3$.
Hard: (a) $(3x-4)(x-2) \Rightarrow x = \frac{4}{3}, 2$ (b) $(2x+1)(x-3) \Rightarrow x = -\frac{1}{2}, 3$ (c) $(3x+2)(2x-1) \Rightarrow x = -\frac{2}{3}, \frac{1}{2}$ (d) $(2x-3)(2x+1) \Rightarrow x = \frac{3}{2}, -\frac{1}{2}$.

3. Quadratic Formula

(a) $x = \dfrac{-3 \pm \sqrt{49}}{2} = \dfrac{-3 \pm 7}{2} \Rightarrow x = 2, -5.$ (b) $x = \dfrac{2 \pm \sqrt{36}}{2} = \dfrac{2 \pm 6}{2} \Rightarrow x = 4, -2.$
(c) $x = \dfrac{-5 \pm \sqrt{49}}{4} = \dfrac{-5 \pm 7}{4} \Rightarrow x = \frac{1}{2}, -3.$ (d) $x = \dfrac{4 \pm \sqrt{12}}{2} = 2 \pm \sqrt{3}$ (about 3.73 and 0.27).

4. Radical Functions

Square root (range is $y \geq 0$ for a–c):
(a) $x \geq 3$, $[3, \infty)$; $y \geq 0$. (b) $x \geq -5$, $[-5, \infty)$; $y \geq 0$. (c) $2x - 4 \geq 0 \Rightarrow x \geq 2$; $y \geq 0$. (d) $x \geq 0$; range $y \geq 2$ (the $+2$ lifts it).
Cube root (all reals for both, in every case):
(a)–(d) Domain: all reals $(-\infty, \infty)$; Range: all reals $(-\infty, \infty)$.

5. Geometry

Interior angles: (a) 720° (b) 1080° (c) 360° (d) 1440°.
Classify: (a) all sides = 3 with right angles ⇒ **square**. (b) sides 5 and 2 with right angles ⇒ **rectangle**.
(c) all sides = $\sqrt{13}$, no right angle ⇒ **rhombus**. (d) opposite sides equal/parallel ($AB = DC = (4, 0)$, $AD = BC = (2, 3)$), not equal adjacent, no right angle ⇒ **parallelogram**.
Cylinder surface area: (a) $32\pi \approx 100.5$ (b) $150\pi \approx 471.2$ (c) $88\pi \approx 276.5$ (d) $18\pi \approx 56.5$ square units.

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